MANSI VEDPATHAK

AI & Data Science Student | Aspiring AI/ML Engineer
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Open to part-time internships • Remote • Flexible hours

SUMMARY

Third-year AI & Data Science undergraduate who builds and deploys full-stack AI applications (multi-agent LLM systems, FastAPI backends, React/Next.js frontends) and measures whether they actually work, using classifiers, detectors, and test suites rather than vibes. Interested in AI/ML and Generative AI engineering internships.

PROJECTS

Other Minds — Multi-Agent LLM Conversation App | GitHub | Live Demo **2026**

Personal Project | Python, FastAPI, Next.js/React, Streamlit, LangChain, Gemini API, SQLite, Docker

- Architected a 3-agent conversational AI system where independent agents answer from distinct psychological worldviews, using isolated system-prompt conditioning so perspectives stay genuinely unreconciled instead of collapsing into one helpful tone
- Designed a single server-side turn orchestrator (FastAPI) that decides who speaks next, consumed by independent Next.js and Streamlit clients plus a seed script, keeping conversation logic consistent across every interface
- Validated agent distinctiveness with a TF-IDF + Logistic Regression classifier under 5-fold cross-validation: 75.3% accuracy vs a 33.3% chance baseline; iterated through 3 prompt designs to eliminate uniform "essay-length" replies
- Built an epistemics detector enforcing a "hypothesis is not fact" rule: grades claims about a user's inner state on a six-level scale (reported to invented). Passes 100/100 hand-written spec cases and 16/25 adversarial cases, with all 9 gaps documented; backend suite of 263 passing tests
- Built an analytics pipeline (Pandas, SQLite/WAL) with idempotent writes and append-only telemetry for reproducible data; deployed as 3 containerized services via Docker Compose (API on Railway, frontend on Vercel, secrets injected at runtime, never baked into images)

Branch Wars — 2048-Style Engineering Meme Game | GitHub | Live Demo **2026**

Personal Project | React, Vite, JavaScript, Web Audio API, Vitest, Python

- Directed end-to-end development of a 2048-style tile-merging browser game, defining mechanics, UI flow, architecture, and features from concept to shipped product with AI-assisted development; themed around Indian engineering branches (Civil to CS) with custom college events, animations, and live-synthesized sound effects (no audio files)
- Integrated an expectimax search AI (hint + autoplay) with board-fullness-adaptive search depth and a four-heuristic evaluator (empty cells, monotonicity, smoothness, corner anchoring), adapted to the game's single-value tile spawns
- Ran a standalone Python experiment (self-play simulation, hill-climbing search, held-out evaluation) to empirically tune the solver's heuristic weights
- Reviewed and iterated on the generated architecture: framework-agnostic game and AI logic kept as pure functions separate from React UI, state centralized in one custom hook, unit-tested with Vitest across 32 commits

CERTIFICATIONS

- Machine Learning with Python — IBM (Coursera), May 2026

EDUCATION

B.Tech, Artificial Intelligence & Data Science — D. Y. Patil College of Engineering, Akurdi, Pune 2024 – 2028

TECHNICAL SKILLS

- **Languages** — Python, JavaScript, TypeScript
- **ML & Data Science** — NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Feature Engineering, Model Evaluation (cross-validation), Statistical Analysis, Data Visualization
- **Generative AI** — LLM APIs (Gemini, OpenAI, Claude), LangChain, Microsoft AutoGen, Prompt Engineering, RAG Systems, Embeddings, Vector Search (Pinecone, HNSW), Search Algorithms (Expectimax)
- **Frontend** — React, Next.js, Vite, TypeScript, Tailwind CSS, Redux, Streamlit
- **Backend** — FastAPI, Node.js, REST APIs, GraphQL
- **Databases & Cloud** — SQLite, MySQL, PostgreSQL, MongoDB, Redis, BigQuery, AWS, Vercel, Railway
- **Tools** — Git, GitHub, GitHub Actions, Docker, Docker Compose, Vitest